

Validation scorecard: twelve executed experiments

E1	Leave-one-city-out CV <i>[Calibration]</i>	ρ_{LOO} value = 0.52 threshold = 0.30	CLEAN
E2	Weight-space LOO bootstrap <i>[Calibration]</i>	Top-10 retention value = 0.90 threshold = 0.50	CLEAN
E3	Behavioural vs eco-counters <i>[Construct]</i>	Partial R^2 net size value = 0.24 threshold = 0.10	CLEAN
E4	Exposure-adjusted safety (city-level proxy) <i>[Construct]</i>	ρ_S shift (raw $\rightarrow$ adj.) value = 0.92 threshold = 0.50	SUBSTITUTE
E5	Within-city station bootstrap (radius proxy) <i>[Geometry]</i>	Kendall τ (median) value = 0.85 threshold = 0.70	SUBSTITUTE
E6	IES specification sweep <i>[Equity]</i>	Robust deserts $P \geq 0.75$ value = 0.83 threshold = 0.50	CLEAN
E7	Sobol/Jansen variance decomposition <i>[Geometry]</i>	$\sum_k S_k$ (score) value = 0.98 threshold = 0.80	CLEAN
E8	Spatial autocorrelation <i>[Panel]</i>	Moran I value = 0.03 threshold = 0.10	CLEAN
E9	Bayesian IES (with prior sensitivity) <i>[Equity]</i>	P-invariant deserts at $\tau \in \{0.1, 1, 10\}$ value = 4.00 threshold = 3.00	QUALIFIED
E10	Per-city IMD bootstrap CI <i>[Panel]</i>	Top-3 CI-distinct value = 1.00 threshold = 0.80	CLEAN
E11	Parametric buffer-radius sweep (lit. elasticities) <i>[Geometry]</i>	robust Kendall τ (150-500m) value = 0.92 threshold = 0.70	SUBSTITUTE
E12	Cook-style leverage (Lyon-removed) <i>[Panel]</i>	$\tau(\text{ref}, -\text{Lyon})$ value = 0.83 threshold = 0.70	QUALIFIED
E13	k-means typology <i>[Panel]</i>	Silhouette ($k = 4$) value = 0.30 threshold = 0.20	CLEAN